Sakshi Sharma

AIML & Generative AI Engineer

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Professional Summary

AI/ML and Generative AI Engineer and final-year B.Tech Data Science student (CGPA 9+) with proven experience designing and deploying LLM-powered applications, RAG pipelines, and autonomous agentic systems. Built production-grade AI products including a multi-step agentic job application pipeline (ReAct-style tool-use), a RAG-based legal assistant and an AI carbon credit platform. Proficient in LangChain, FAISS, Hugging Face, prompt engineering, fine-tuning (PEFT, SFT), and agentic workflow design with LangGraph/AutoGen. SIH Finalist, THDCIL intern, and government-funded researcher; seeking to build next-generation AI products powered by foundation models.

Education

B.Tech in Data Science — IPS Academy IES, Indore *2023 – 2027 | CGPA: 9+*
12th Grade — Karnataka Vidya Niketan *2022*

Work Experience

AI/ML Backend Engineer (Intern) – THDCIL – Ministry of Power *Sep 2025 – Present*
- Engineered an automated document verification pipeline using hash-based tamper detection, reducing manual review effort by ~60%
- Implemented forgery detection models integrated into secure backend workflows; pipeline processes 200+ documents per day with sub-second latency.
- Stack: Python, FastAPI, PostgreSQL, ML classification models

ML Systems Developer – Autonomous Vehicle for Calamity Rescue (AVCR) – Funded Research Project *2025 – Present*
- Deployed ML inference pipeline on NVIDIA Jetson Nano for real-time survivability assessment in disaster-hit confined spaces
- Achieved 91% classification accuracy using an ensemble of Isolation Forest, LSTM, and Random Forest models on live sensor streams
- Project received institutional funding; selected among top research projects at IPS Academy IES

Data Analytics Virtual Intern – Deloitte Australia (Forage) *Apr 2025*
- Built a Tableau dashboard for forensic data analysis; applied Excel-based classification to surface business insights from structured datasets

Projects

Auto Mail Apply — Agentic AI Job Application Pipeline
- Designed and built a fully autonomous multi-step agent that scrapes job listings, ranks candidate-job fit using NLP (cosine similarity + keyword extraction), and auto-composes and sends tailored outreach emails with zero manual input
- Implemented tool-use architecture: agent invokes web scraper, profile matcher, and email composer as sequential tools — a working implementation of ReAct-style agentic workflow
- Reduced time-per-application from ~20 minutes to under 30 seconds; scales across multiple job boards and candidate profiles
- Stack: Python, FastAPI, Selenium, spaCy, LLM tool-calling

AI-Powered Personal Legal Assistant (RAG + LLM)
- Built a production-grade RAG pipeline combining FAISS vector retrieval with LLM generation for legal document summarization, case strategy advisory, and document classification
- Reduced LLM hallucination rate by ~35% vs. baseline through retrieval-augmented grounding; supports multi-document context windows up to 32k tokens
- Implemented multi-turn conversational memory and streaming responses for real-time UX
- Stack: FastAPI, FAISS, LangChain, Groq/OpenAI APIs, Next.js 14, PostgreSQL

CarbonVault — AI Carbon Credit Platform
- Built an end-to-end AI platform automating carbon credit issuance for NGOs, admins, and corporates across 500+ submission workflows — integrating satellite imagery (NDVI analysis) for automated land cover validation
- Reduced manual verification turnaround from days to under 2 minutes; role-based audit workflows with multi-user auth
- Stack: FastAPI, React, SpacetimeDB, PostgreSQL, Vercel

Rockfall Prediction & Alert System
- ML classification system predicting rockfall risk in open-pit mines from geological + vibration sensor data; 88% precision on held-out test set
- Real-time alerting pipeline with 10–15 minute early warning ahead of critical risk threshold

Skills

Programming Languages: Python, SQL, C++, C
Machine Learning & Deep Learning: Supervised and Unsupervised Learning, Feature Engineering, Scikit-learn, XGBoost, Deep Learning, CNNs, RNNs, LSTMs, Transformers, Attention Mechanisms, Reinforcement Learning, PyTorch, TensorFlow, Scikit-learn, ML flow
Generative AI & LLMs: GPT-4, Claude, Gemini, LLaMA, Hugging Face Transformers, Prompt Engineering, Few-Shot Prompting, Chain-of-Thought Prompting, Instruction Tuning, Fine-Tuning (PEFT, SFT), RLHF (Conceptual)
Agentic AI & RAG: LangChain, LangGraph, LlamaIndex, Retrieval-Augmented Generation (RAG), ReAct Agents, Multi-Agent Systems, Tool Calling, Function Calling, Autonomous Workflows, Task Planning and Decomposition
Vector Databases & Search: ChromaDB, FAISS, Embedding Models (sentence-transformers, OpenAI Embeddings), Hybrid Search, Semantic Chunking
Computer Vision & Multimodal AI: OpenCV, YOLOv8, MediaPipe, Image Segmentation, Object Detection, Vision-Language Models
Backend & Deployment: FastAPI, Django, WebSockets, Docker, MLflow, CI/CD, Model Serving
Databases & Analytics: PostgreSQL, MySQL, Pandas, NumPy, Matplotlib, Exploratory Data Analysis (EDA), Power BI
Cloud & Tools: AWS (EC2, S3), Git, GitHub, Vercel, Google Colab, VS Code
Soft Skills: Problem Solving, Research-to-Product Execution, Rapid Prototyping, Technical Presentations, Hackathon Leadership

Achievements

Winner – UDAAN'26 Phase I and Phase II | Techno Maths Presentation | Science Exhibition 2024 | VyaparX Idea Pitching Competition | Inter-College Chess Competition 2023, 25

Smart India Hackathon (SIH) Finalist | NEC (National Entrepreneurship Challenge) Finalist 2024'25 | Top 20- Clear Skies Innovation Challenge | Top 100 Campus Ambassador, IIT Bombay

Participations hackathons – Bhartiya Antariksha Hackathon, Hacksungum, GDSC Codespire, Pravah 3.0, Hackdata' 25, Build with India, Samsung Solve for Tomorrow, Flipkart Grid 6.0 (Robotics), Snow Hacks

Certifications

- Artificial Intelligence and Machine Learning — NPTEL
- AWS Cloud Operations — Training Badge
- RDBMS PostgreSQL
- Python Tutorial — IIT Bombay

Leadership

- **Founder, E-Cell** — IES IPS Academy: Established the entrepreneurship cell; organized pitching events, mentorship sessions, and cross-college idea competitions